

B.Tech. Degree VI Semester Examination April 2013**ME 604 HEAT AND MASS TRANSFER**
(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A
(Answer **ALL** questions)

(8 × 5 = 40)

- I. (a) What is the critical radius of insulation? Consider an insulated pipe exposed to the atmosphere. Will the critical radius of insulation be greater on calm days or on windy days? Why?
- (b) What is lumped system analysis? For which solid is the lumped system analysis more likely to be applicable: an actual apple or a golden apple of the same size? Why?
- (c) Define the Prandtl number. What is the physical significance of the Prandtl number?
- (d) Draw the boiling curve and identify the different boiling regimes. Also, explain the characteristics of each regime.
- (e) Explain:
 (i) Wien's displacement law
 (ii) Stefan-Boltzmann law
- (f) Consider a hemispherical furnace with a flat circular base of diameter D . Determine the view factor from the dome of this furnace to its base.
- (g) Represent the variation of fluid temperature in a heat exchanger when one of the fluids condenses or boils. Why do the directions of flow is immaterial in such cases?
- (h) Explain the effectiveness and NTU of a heat exchanger.

PART B

- II. (a) Starting with energy balance on a volume element, derive the general heat conduction equation in rectangular coordinates. (9)
- (b) A current of 200A is passed through a stainless-steel wire $[k = 19 \text{ W/m}^\circ\text{C}]$ 3mm in diameter. The resistivity of the steel may be taken as $70 \mu\Omega \text{ cm}$ and the length of the wire 1m. The wire is submerged in a liquid at 110°C and experiences a convection heat-transfer coefficient of $4 \text{ kW/m}^2 \text{ }^\circ\text{C}$. Calculate the center temperature of the wire. (6)
- OR**
- III. (a) Steam in a heating system flows through tubes whose outer diameter is 3cm and whose walls are maintained at a temperature of 120°C . Circular aluminium fins $(k = 180 \text{ W/m}^\circ\text{C})$ of outer diameter 6cm and constant thickness 2mm are attached to the tube. The space between the fins is 3mm, and thus there are 200 fins per meter length of the tube. Heat is transferred to the surrounding air at 25°C , with a combined heat transfer coefficient of $60 \text{ W/m}^2 \text{ }^\circ\text{C}$. Determine the increase in heat transfer from the tube per meter of its length as a result of adding fins. (9)
- (b) The ground at a particular location is covered with snow pack at 8°C for a continuous period of 60 days, and the average soil properties at that location are $k = 0.35 \text{ W/m}^\circ\text{C}$ and $\alpha = 0.15 \times 10^{-6} \text{ m}^2/\text{s}$. Assuming an initial uniform temperature of -8°C in the ground, determine the minimum burial depth to prevent the water pipes from freezing. (6)

(P.T.O.)

- IV. (a) Air at 27°C and 1atm flows over a flat plate at a speed of 2 m/s. Calculate the boundary layer thickness at distances of 20cm and 40cm from the leading edge of the plate. Calculate the mass flow that enters the boundary layer between $x = 20\text{cm}$ and $x = 40\text{cm}$. The viscosity of air at 27°C is $1.85 \times 10^{-5} \text{ Kg/ms}$. Assume unit depth in the z direction. (7)

- (b) Hot air flows with a mass rate of $m = 0.050 \text{ Kg/s}$ through an uninsulated sheet metal duct of diameter $D = 0.15\text{m}$, which is in the crawlspace of a house. The hot air enters at 103°C and, after a distance of $L = 5\text{m}$, cools to 85°C . The heat transfer coefficient between the duct outer surface and the ambient air at $T_{\infty} = 0^{\circ}\text{C}$ is known to be $h_o = 6 \text{ W/m}^2 \text{ K}$. (8)

- (i) Calculate the heat loss (W) from the duct over the length L .
 (ii) Determine the heat flux and the duct surface temperature at $x = L$.

OR

- V. (a) An electronic circuit board in the shape of a flat plate $0.3 \text{ m} \times 0.3 \text{ m}$ in plan-form and dissipates 15W. It may be placed in operation on an insulated surface in a horizontal position or at an angle of 45 degrees to horizontal, both in still air at 25°C . If the circuit would fail above 60°C , determine if the two proposed installations are safe. Assume that radiative heat transfer is negligible. (9)
- (b) Air at 25°C and 50% RH flows over water surface $12\text{m} \times 6\text{m}$ at a velocity of 2 m/s. Determine the water loss per day considering that flow direction is along the 12m side. Take $D_{AB} = 0.26 \times 10^{-4} \text{ m}^2/\text{s}$, $Sc = 0.60$, $\nu = 15.7 \times 10^{-6} \text{ m}^2/\text{s}$. (6)

- VI. (a) A glass plate of 30cm square is used to view radiation from a furnace. The transmissivity of the glass is 0.5 from 0.2 to $3.5\mu\text{m}$. The emissivity may be assumed to be 0.3 upto $3.5\mu\text{m}$ and 0.9 above that. The transmissivity of the glass is zero, except in the range from 0.2 to $3.5\mu\text{m}$. Assuming that the furnace is a blackbody at 2000°C , calculate the energy absorbed in the glass and the energy transmitted. (7)

- (b) A furnace is in the shape of a cylinder of 1m diameter and 1m height. The base is at 1000K while the top is at 500K. The curved surface is well insulated so that it can be taken as non absorbing reradiating surface. Determine the heat exchange between the base and the top with and without the reradiating surface. (8)

OR

- VII. (a) State and prove Kirchhoff's law for heat radiation. (6)
- (b) A cryogenic fluid flow through a long tube of 20mm diameter, the outer surface of which is diffuse and gray with $\epsilon_1 = 0.02$ and $T_1 = 77\text{K}$. This tube is concentric with a larger tube of 50mm diameter, the inner surface of which is diffuse and gray with $\epsilon_2 = 0.05$ and $T_2 = 300\text{K}$. The space between the surfaces is evacuated. Calculate the heat gain by the cryogenic fluid per unit length of tubes. If a thin radiation shield of 35mm diameter and $\epsilon_3 = 0.02$ (both sides) is inserted midway between the inner and outer surfaces, calculate the change (percentage) in heat gain per unit length of the tubes. (9)

- VIII. (a) Derive an expression for the mean temperature difference in a counter-flow heat exchanger. (7)

- (b) A counter flow, concentric tube heat exchanger is used to cool the lubricating oil for a large industrial gas turbine engine. The flow rate of cooling water through the inner tube ($D_1 = 25\text{mm}$) is 0.2 Kg/s, while the flow rate of oil through the outer annulus ($D_0 = 45\text{mm}$) is 0.1 Kg/s. The oil and water enter at temperatures of 100 and 30°C , respectively. How long must the tube be made if the outlet temperature of the oil is to be 60°C ? (8)

OR

- IX. (a) Derive the expression for the effectiveness of a parallel flow heat exchanger. (7)
- (b) Exhaust gases from a power plant are used to preheat air in a cross-flow heat exchanger. The exhaust gases enter the heat exchanger at 450°C and leave at 200°C . The air enters the heat exchanger at 70°C , leaves at 250°C , and has a mass flow rate of 10Kg/s. Assume that properties of the exhaust gases can be approximated by those of air. The overall heat transfer coefficient of the heat exchanger is $154 \text{ W/m}^2 \text{ K}$. Calculate the heat exchanger surface area required if (i) the air is unmixed and the exhaust gases are mixed and (ii) both fluids are unmixed. (8)